

Genetic report for longevity

Extracellular Matrix and Glycation

Collagen, elastin, metalloproteinases, AGEs/RAGE, fibrosis and vessel stiffness

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The extracellular matrix (collagen, elastin, proteoglycans) gives structure to the skin, vessels, tendons, lungs and bone. With age, the matrix is degraded by metalloproteinases, becomes more rigid by cross-linking and by advanced glycation end products (AGEs), and elastin is fragmented without being restored. This report reads the genes of these processes.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

31

variants

22

in the chip

8

imputed
($DR2 \geq 0.8$)

1

not found

8

Scores with
percentile

Overall reading. The matrix is genetically well protected in glycation (HbA1c genetics at the 7th percentile; AGER without the 82S allele) and arterial stiffness (score at the 12th percentile), and the spine also (disc hernia at the 3rd percentile). The clear weak point is elastin: the Pi*Z allele of alpha-1 antitrypsin in heterozygosity (PiMZ) leaves less protection against neutrophil elastase in the lung, skin and vessels, and is also a cofactor of hepatic fibrosis. MUC5B does not have the pulmonary fibrosis allele and LOXL1 G153D is heterozygous (less risk of pseudoexfoliation than the average). Inguinal hernia, varices, Dupuytren and aneurysm of the aorta are typical. The remaining unfavourable signals (COL5A1, LOXL1 R141L, MMP9, FBN1) are of weak evidence.

Favourable

- **Genetic glyated hemoglobin (HbA1c):** the 7th percentile
- **Low back disc hernia:** the 3rd percentile
- **Arterial stiffness (index of stiffness and pulse velocity):** the 12th percentile
- **SERPINA1 rs17580 TT:** Pi*S (E264V): Allele A gives ~60% of antitrypsin per copy; effect smaller than Z, but adds to it (SZ).
- **LOXL1 rs3825942 AG:** G153D: the G allele (very common) is strongly associated with pseudoexfoliation syndrome (accumulation of elastic fibrillar material in the eye, cause of glaucoma) and pelvic prolapse
- **MUC5B rs35705950 GG:** MUC5B promoter: T allele increases mucin 5B in bronchioles and is the highest common genetic risk factor of idiopathic pulmonary fibrosis (OR ~5 per copy).
- **EFEMP1 rs3791679 GA:** Fibulina-3: elastic matrix protein; hernia locus.
- **WT1 rs2301250 AA:** Locus WT1 of inguinal hernia.

Unfavourable

- **SERPINA1 rs28929474 CT:** Pi*Z (E342K): the T allele forms antitrypsin alpha-1 polymers that are retained in the liver; the lung and skin are less protected against neutrophil elastase

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 22 chip genotypes, 0 needed complement, 1 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Collagen and connective tissue integrity

Fibrillary collagens (types I, III and V) and polygenic scores of diseases in which the connective tissue gives rise: inguinal hernia, varices, disc hernia and Dupuytren's disease (palmar fibromatosis, the opposite example: excess matrix).

Ageing mechanisms: Remodelling of the extracellular matrix; Stiffness of the extracellular matrix; Fragility

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Inguinal hernia	Typical / intermediate	the 65th percentile · z=0.38	Higher score : more risk (weakness of the abdominal wall and collagen). PGS001854 (PGS Catalog; Penalized regression; Inguinal hernia; PGP000263, PMID 34995502). Variants with genotype: 7127 of 7326 (covering of 97.3% coefficients).
Variations	Typical / intermediate	the 64th percentile · z=0.33	Higher score : more risk (vein wall and valves). PGS001845 (PGS Catalog; Penalized regression; Varicose vein; PGP000263, PMID 34995502). Variants with genotype: 15341 of 15748 (covering of 96.8% coefficients).
Dupuytren disease (palmar fibromatosis)	Typical / intermediate	the 38th percentile · z=-0.35	Higher score : higher risk of excess matrix in palmar fascia. PGS001880 (PGS Catalog; Penalized regression; Contracture of palmar fascia [Dupuytren's disease]; PGP000263, PMID 34995502). Variants with genotype: 2361 of 2443 (covering of 96.9% coefficients).
Low back disc hernia	Favourable	the 3rd percentile · z=-1.58	Higher score : higher risk. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 57 of 62 loci with genotype. Studies: Salo V (Lumbar disc herniation requirement); Salo V (Lumbar disc herniation). Variants with genotype: 57 of 62 (covering of 85.1% coefficients).

Favourable

- Lumbar disc hernia (the 3rd percentile)
- COL3A1 rs1800255 GG
- EFEMP1 rs3791679 GA
- WT1 rs2301250 AA
- ADAMTS6 rs6892539 CC

Unfavourable

- COL5A1 rs12722 TT

Favourable variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL3A1	rs1800255 2:188999354 · G/A/C/T	GG	chip	A (0; 26%)	weak	A698T: Allele A (698T) was associated with prolapse of pelvic organs, hernias and aneurysms in small studies. The collagen type III is the collagen of vessels and young skin.
EFEMP1	rs3791679 2:55869757 · A/C/G/T	GA	chip	A (1; 77%)	moderate	Fibulina-3: elastic matrix protein; hernia locus. Meaning: GWAS Catalog: Abdominal hernia (PheCode 550) (Verma A 2024; $p=3e-81$); Allele A ↑
WT1	rs2301250 11:32436673 · G/A	AA	imputation DR2=0.99	G (0; 65%)	moderate	Inguinal hernia WT1 locus. Meaning: GWAS Catalog: Abdominal hernia (PheCode 550) (Verma A 2024; $p=2e-60$); G ↑ allele
ADAMTS6	rs6892539 5:65126255 · C/G/T	CC	imputation DR2=0.97	T (0; 69%)	weak	ADAMTS6: matrix protease (microfibrils). Meaning: GWAS Catalog: Inguinal hernia (Wei J 2021; $p=7e-14$); T ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL5A1	rs12722 9:134842570 · C/T	TT	chip	T (2; 58%)	weak	BstUI (3'UTR): T allele is associated with tendinopathy and ligament injuries in small studies. Type V collagen regulates the diameter of fibrils.

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL1A1	rs1800012 17:50200388 · C/A	CC	chip	A (0; 19%)	moderate	Sp1: the "s" allele (A in strand +) alters the proportion of collagen type I chains; it is associated with bone and more fragile connective tissue (GENOMOS).
COL1A2	rs17166182 7:94330606 · C/A	AC	imputation DR2=0.99	C (1; 67%)	weak	Type I $\alpha 2$ collagen (main variant of GWAS). Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=3e-11); C $\uparrow$ allele

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL1A2	rs42524 7:94413927 · C/A/G/T	GG	chip	—	weak	P549A: common variant of collagen type I $\alpha 2$; small and unreplicated associations (intracranial aneurysms, prolapse). Information.

Elastin and elastic fibers

Elastin is made almost only in childhood and adolescence and lasts for decades; what is lost then is hardly restored. Neutrophil elastase protection (alpha-1 antitrypsin, SERPINA1) and crosslinking by LOX/LOXL1 are key points.

Ageing mechanisms: Fragmentation of Elastin; Vascular Aging; Arterial stiffness; Remodelling of the extracellular matrix

Favourable

- SERPINA1 rs17580 TT
- LOXL1 rs3825942 AG
- ELN rs17855988 CG

Unfavourable

- SERPINA1 rs28929474 CT
- LOXL1 rs1048661 GG
- FBN1 rs1036476 TC

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SERPINA1	rs17580 14:94380925 · T/A	TT	chip	A (0; 6%)	moderate	Pi*S (E264V): Allele A gives ~60% of antitrypsin per copy; effect smaller than Z, but adds to it (SZ).
LOXL1	rs3825942 15:73927241 · G/A/C/T	AG	chip	G (1; 82%)	moderate	G153D: the G allele (very common) is strongly associated with pseudoexfoliation syndrome (accumulation of elastic fibrillary material in the eye, cause of glaucoma) and pelvic prolapse. LOXL1 reticulates to elastin.
ELN	rs17855988 7:74060495 · G/A/C	CG	imputation DR2=0.98	G (1; 91%)	weak	Elastin (main variant of GWAS). Meaning: GWAS Catalog: Hernia (any subtypes) (Ahmed WU 2022; p=4e-12); G ↑ allele

Unfavourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SERPINA1	rs28929474 14:94378610 · C/G/T	CT	chip	T (1; 2%)	strong	Pi*Z (E342K): the T allele forms antitrypsin alpha-1 polymers that are retained in the liver; the lung and skin are less protected against neutrophil elastase. 1 copy (PiMZ) = ~60% of normal antitrypsin; with tobacco or pollution, plus emphysema; also liver fibrosis cofactor.
LOXL1	rs1048661 15:73927205 · G/A/T	GG	chip	G (2; 68%)	weak	R141L: the G allele was associated with pseudoexfoliation in several studies (the sense is reversed in some populations).
FBN1	rs1036476 15:48622578 · T/C	TC	imputation DR2=1.00	C (1; 11%)	weak	Fibrillin-1 (microfibrils; Marfan syndrome in rare forms). Meaning: GWAS Catalog: Aortic aneurysm (Sakaue S 2021; p=9e-10); C ↑ allele

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ELN	rs2071307 7:74056384 · G/A/C	GA	chip	A (1; 42%)	weak	G422S (20 exon): allele A was associated with intracranial aneurysms and arterial stiffness in small studies; not confirmed in large GWAS.

Metalloproteinases and remodeling

Matrix metalloproteinases (MMP-1, -2, -3, -9, -12) degrade collagen and elastin; rise with UV radiation, inflammation and tobacco. The lysyl oxidase (LOX) makes the collagen and elastin normal crosslinks (copper need).

Ageing mechanisms: Remodelling of the extracellular matrix; Chronic Inflammation; Vascular Aging

Favourable

None.

Unfavourable

- MMP9 rs3918242 CT

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MMP9	rs3918242 20:46007337 · C/T	CT	imputation DR2=0.99	T (1; 17%)	weak	-1562C>T: T allele increases the transcription of MMP-9 (gelatinase B); associated with aneurysms and arterial stiffness in small studies.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MMP3	rs679620 11:102842889 · T/A/C/G	CC	chip	—	weak	E45K: in LD with the 5A/6A of the promoter (unread indel). Associations with tendinopathy and inconsistent coronary disease. Information.
MMP2	rs243865 16:55477894 · C/T	CC	chip	—	weak	-1306C>T: T allele reduces the activity of the MMP-2 promoter; clinical effects in opposite directions according to the tissue. Information.
MMP12	rs2276109 11:102875061 · T/A/C/G	CT	chip	—	weak	-82A>G: G allele reduces the expression of MMP-12 (elastase of macrophages) and is associated with better lung function in smokers. Information.
LOX	rs1800449 5:122077513 · C/A/T	CC	chip	—	weak	R158Q (in propeptide): allele A was associated with lower activity of lysyl oxidase and aneurysms in small studies. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MMP1	rs1799750 11:102799765 · CC/C	—	SNP not found in the data	—	moderate	-1607 1G/2G (indel no promoter): the 2G allele creates a ETS site and increases the expression of MMP-1 (interstitial collagenase); it is an insertion/deletion, almost never read in the chip.

Glycation, AGEs and RAGE

The AGEs form when sugars and methylglyoxal react with long-lived proteins (collagen, elastin, crystalline). Glyoxalase 1 (GLO1) eliminates methylglyoxal, fructosamina-3-cinase (FN3K) undoes initial glycation and the RAGE receptor (AGER gene) transforms AGEs into an inflammatory signal; the soluble form (sRAGE) functions as bait.

Ageing mechanisms: Accumulation of Advanced Glycation Final Products (AGEs); Hyperglycemia and Intolerance to Glucose; Stiffness of the extracellular matrix; Inflammaging

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Genetic glycated hemoglobin (HbA1c)	Favourable	the 7th percentile · z=-1.51	Higher score = higher HbA1c (more glycation throughout life). Part of the signal comes from the biology of the red blood cell and not from glucose. PGS001953 (PGS Catalog; Penalized regression; Glycated haemoglobin (HbA1c); PGP000263, PMID 34995502). Variants with genotype: 44856 of 46566 (covering of 96.2% coefficients).
low circulating sRAGE (AGEs bait)	Typical / intermediate	the 31st percentile · z=-0.55	Higher score = genetically lower sRAGE (less bait for AGEs). Few loci; dominated by AGER. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 3 of 3 loci with genotype. Studies: Grauen Larsen H (Soluble receptor for advanced glycation end-product levels); Maruthur NM (Soluble receptor for advanced glycation end-product levels). Variants with genotype: 3 of 3 (covering of 100.0% coefficients).

Favourable

- Genetic glycated hemoglobin (HbA1c) (the 7th percentile)

Unfavourable

None.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
AGER	rs2070600 6:32183666 · C/T	CC	chip	T (0; 5%)	moderate	G82S: the T allele (82S) strongly lowers the circulating sRAGE (the bait that neutralizes AGEs) and increases the affinity of the receptor to ligands; it is associated with worse lung function and more inflammation.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GLO1	rs4746 6:38682852 T/A/G	GG	chip	—	weak	E111A: common variant of Glyoxalase 1; the effect on enzymatic activity is contradictory between studies. The variation in the number of copies of GLO1 (most important) is not seen on the chip. Information.
FN3K	rs1056534 17:82750725 C/A/G/T	GG	imputation DR2=0.98	—	weak	Fructosamina-3-cinase (declication): a common variant associated with FN3K activity and HbA1c in small studies. Information.
FN3K	rs3848403 17:82736023 C/G/T	CT	imputation DR2=1.00	—	weak	FN3K promoter: was associated with enzyme expression and hemoglobin glycation in small studies. Information.
AKR1B1	rs759853 7:134459206 G/A/T	AG	chip	—	weak	Aldose reductase (polyol pathway, -106C>T): associated with microvascular

Fibrosis

Fibrosis is the excess of matrix after repeated injury (lung, liver, skin, heart). Genes: MUC5B and TOLLIP (lung), TGF-β1 (central fibrotic signal), SERPINA1 (liver) and the idiopathic pulmonary fibrosis score.

Ageing mechanisms: Remodelling of the extracellular matrix; Cell Senescence; Chronic Inflammation; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Idiopathic pulmonary fibrosis (PGS004695)	Typical / intermediate	the 30th percentile	23 variant score, including MUC5B. Source: §18/§20 profile, imputed genome.

Favourable

- MUC5B rs35705950 GG

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MUC5B	rs35705950 11:1219991 · G/A/T	GG	chip	T (0; 11%)	strong	MUC5B promoter: T allele increases mucin 5B in bronchioles and is the highest common genetic risk factor of idiopathic pulmonary fibrosis (OR ~5 per copy).

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TOLLIP	rs5743890 11:1304599 · T/C	TT	chip	—	moderate	TOLLIP: in the GWAS of idiopathic pulmonary fibrosis (Noth 2013), the minor allele (C) was associated with less risk of disease but worse survival in patients; therefore, without judgment.
TGFB1	rs1800470 19:41353016 · G/A/C	AG	chip	—	weak	L10P (10 codon): the Pro allele (G in strand +) was associated with more secretion of TGF-β1; meaning of the effect on fibrosis varies between studies. Information.

Arterial stiffness and aorta

The stiffness of the large arteries (pulse wave velocity) is one of the strongest markers of vascular aging. Scores of arterial stiffness and abdominal aortic aneurysm.

Ageing mechanisms: Arterial stiffness; Vascular Aging; Fragmentation of Elastin; Stiffness of the Extracellular Matrix

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Abdominal aneurysm	Typical / intermediate	the 26th percentile · z=-0.64	Higher score : higher risk. PGS000753 (PGS Catalog; Pruning and Thresholding (P+T); Abdominal aortic aneurysm; PGP000159, PMID 32981348). Variants with genotype: 27 of 29 (covering of 91.5% coefficients).
Arterial stiffness (index of stiffness and pulse velocity)	Favourable	the 12th percentile · z=-1.09	Higher score : genetically rigid arteries. Few loci; low reliability. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 18 of 22 loci with genotype. Studies: Fung K (Arterial stiffness index); Rhode M (Brachial-ankle pulse velocity) Rode M (Brachial-femoral pulse velocity) Rode M (Carotid-femoral pulse velocity) Tarasov KV (Arterial stiffness). Variants with genotype: 18 of 22 (covering of 81.8% coefficients).

Favourable

- Arterial stiffness (index of stiffness and pulse wave velocity) (the 12th percentile)
- COL4A1 rs3742207 TT

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL4A1	rs3742207 · 13:110166251 · T/A/G	TT	chip	G (0; 32%)	weak	IV type collagen (basal vessel membrane). Meaning: GWAS Catalog: Arterial stiffness (Tarasov KV 2009; p=5e-08); G ↑ allele

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LRP1	rs11172113 · 12:57133500 · T/C/G	CT	chip	T (1; 62%)	moderate	LRP1: Aortic aneurysm locus. Meaning: GWAS Catalog: Aortic aneurysm (PheCode 442.1) (Verma A 2024; p=4e-20); allele T ↑

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Remodelling of the extracellular matrix	COL5A1 TT; SERPINA1 CT; LOXL1 GG; FBN1 TC; MMP9 CT	Lumbar disc hernia (the 3rd percentile); COL3A1 GG; EFEMP1 GA; WT1 AA; ADAMTS6 CC; SERPINA1 TT; LOXL1 AG; ELN CG; MUC5B GG	Inguinal hernia (the 65th percentile); Varieties (the 64th percentile); Dupuytren's disease (palmar fibromatosis) (the 38th percentile); COL1A1 CC; Idiopathic pulmonary fibrosis (PGS004695) (the 30th percentile)
Vascular Aging	SERPINA1 CT; LOXL1 GG; FBN1 TC; MMP9 CT	SERPINA1 TT; LOXL1 AG; ELN CG; Arterial stiffness (index of stiffness and pulse wave velocity) (the 12th percentile); COL4A1 TT	Abdominal aneurysm (the 26th percentile); LRP1 CT
Fragmentation of Elastin	SERPINA1 CT; LOXL1 GG; FBN1 TC	SERPINA1 TT; LOXL1 AG; ELN CG; Arterial stiffness (index of stiffness and pulse wave velocity) (the 12th percentile); COL4A1 TT	Abdominal aneurysm (the 26th percentile); LRP1 CT
Arterial stiffness	SERPINA1 CT; LOXL1 GG; FBN1 TC	SERPINA1 TT; LOXL1 AG; ELN CG; Arterial stiffness (index of stiffness and pulse wave velocity) (the 12th percentile); COL4A1 TT	Abdominal aneurysm (the 26th percentile); LRP1 CT
Fragility	COL5A1 TT	Lumbar disc hernia (the 3rd percentile); COL3A1 GG; EFEMP1 GA; WT1 AA; ADAMTS6 CC	Inguinal hernia (the 65th percentile); Varieties (the 64th percentile); Dupuytren's disease (palmar fibromatosis) (the 38th percentile); COL1A1 CC
Chronic Inflammation	MMP9 CT	MUC5B GG	Idiopathic pulmonary fibrosis (PGS004695) (the 30th percentile)
Stiffness of the Extracellular Matrix	COL5A1 TT	Lumbar disc hernia (the 3rd percentile); COL3A1 GG; EFEMP1 GA; WT1 AA; ADAMTS6 CC; Genetic glycated hemoglobin (HbA1c) (the 7th percentile); Arterial stiffness (index of stiffness and pulse wave velocity) (the 12th percentile); COL4A1 TT	Inguinal hernia (the 65th percentile); Varieties (the 64th percentile); Dupuytren's disease (palmar fibromatosis) (the 38th percentile); COL1A1 CC; low circulating sRAGE (AGEs bait) (the 31st percentile); AGER CC; Abdominal aneurysm (the 26th percentile); LRP1 CT
Accumulation of Advanced Glycation Final Products (AGEs)	—	Genetic glycated hemoglobin (HbA1c) (the 7th percentile)	low circulating sRAGE (AGEs bait) (the 31st percentile); AGER CC
Intercellular Communication	—	MUC5B GG	Idiopathic pulmonary fibrosis (PGS004695) (the 30th percentile)
Hyperglycemia and Intolerance to Glucose	—	Genetic glycated hemoglobin (HbA1c) (the 7th percentile)	low circulating sRAGE (AGEs bait) (the 31st percentile); AGER CC
Inflammaging	—	Genetic glycated hemoglobin (HbA1c) (the 7th percentile)	low circulating sRAGE (AGEs bait) (the 31st percentile); AGER CC
Cell Senescence	—	MUC5B GG	Idiopathic pulmonary fibrosis (PGS004695) (the 30th percentile)

Practical part

Practical ideas taken only from the genetics of this report, crossed with the genetic plan v2 and current medication (isotretinoin, rapamycin 4 mg of 6 in 6 weeks, lithium 5 mg, tadalafil, dapagliflozin, retatrutide, telmisartan, LDN 4.5 mg, melatonin). The current stack remains as it is.

1. Protecting elastin (PiMZ)

Genetics: SERPINA1 Pi*Z heterozygous (strong evidence)

Zero tobacco (including passive and vaporizers) and minimum of smoke, dust and pollution; FFP2 mask in works, powder or smoke from fires. To treat early respiratory infections. Confirm the phenotype Pi and alpha-1 serum antitrypsin, as already indicated in the profile.

V2 genetic plan and medication: Similar to the profile (item 9 and §10.4) and the plan v2 (liver: PNPLA3 + MBOAT7 + Pi*MZ). • **suggestion (not applied)**

2. Low glycation as a habit

Genetics: HbA1c genetic p7 (favourable); AGEs are mainly environmental

Keeping glucose stable (feedbacks with protein, fiber and fat; low glycemic index hydrates). More fiber is always better. Cooking more in wet (cooking, steaming) and less fried, grilled and processed meats, which bring ready AGEs; marinate in acid (lime, vinegar) before grilling reduces the formation of AGEs.

V2 genetic plan and medication: Coherent with the v2 plan (fiber ≥ 50 g/day; dapagliflozin and retatrutide already lower glucose). • **suggestion (not applied)**

3. Antiglycation and collagen in the stack

Genetics: AGER CC; COL1A1 without "s"; COL5A1 TT (weak)

Keep L-carnosine 2 g (antiglycation), collagen 20 g, glycine 5 g and vitamin C between meals. Benfotiamine 150–300 mg/day can reduce the formation of AGEs via transketolase (moderate evidence in short studies).

V2 genetic plan and medication: Carnosine (S21), collagen (S9), glycine (S8) and vitamin C (S16) already in the plan; benfotiamine would be new. • **suggestion (not applied)**

4. Copper sufficient for lysyl oxidase

Genetics: LOX/LOXL1 (reticulation of collagen and elastin)

LOX and LOXL1 need copper. With zinc 20–25 mg/day, maintain copper 1.5–2 mg/day in total, as in the plan.

V2 genetic plan and medication: Similar to the v2 plan (S12: zinc + copper). • **suggestion (not applied)**

5. Vitamin K2 and arteries

Genetics: Arterial stiffness p12 (favourable); MGP depends on vitamin K

Maintain K2 MK-7 240 µg: the matrix Gla protein (MGP) locks the calcification of the arteries.

V2 genetic plan and medication: In the plan (S2, K2 KEEP). • **suggestion (not applied)**

6. Aerobic exercise and heat for vessels

Genetics: Arterial stiffness p12; FBN1 and MMP9 (weak)

Regular aerobic training (3–4x/week, 15–20 min) are associated with less rigid arteries; very heavy and isolated strength training temporarily increases stiffness, so it is convenient to join the aerobic.

V2 genetic plan and medication: Compatible with the plan. • **suggestion (not applied)**

7. Medication that helps the matrix

Genetics: TGF- β /fibrosis; elastin; glycation

Telmisartan: the blockade of AT1 receptor reduces the signaling of TGF- β (base of trials with sartans in aortic diseases) and is anti-fibrotic. Rapamycin: anti-fibrotic and anti-senescence in animal models. Dapagliflozin and retatrutide: less glucose, less AGEs. Tadalafil: improves endothelium function. Isotretinoin: no relevant effect on the matrix at this dose.

V2 genetic plan and medication: No changes to the medication. • **suggestion (not applied)**

8. Monthly senolytic only with normal ALT

Genetics: Senescence and fibrosis (MUC5B GG, PiMZ)

The monthly senolytic pulse (fisetin + quercetin) only with normal ALT; pause curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed).

V2 genetic plan and medication: Similar to the v2 plan aligned with the supplemental protocol. • suggestion (not applied)

Avoiding or with caution

- Tobacco, vapor and passive smoke (PiMZ: pulmonary elastin and cutaneous). suggestion (not applied)
- Excess of sugars and fructose, fried and processed meat (AGEs). suggestion (not applied)
- Fluorquinolones antibiotics if there is an alternative (degrade collagen: tendons and aorta). suggestion (not applied)
- Sun exposure without protection (UV radiation activates MMP-1 and causes solar elastosis). suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. Important indels (MMP1 1G/2G, MMP3 5A/6A) and the number of copies of GLO1 are not read.
- AGEs depend mainly on glucose throughout life, diet and tobacco, not genes. The glycation genes (GLO1, FN3K, AKR1B1) have small and poorly replicated effects.
- Polygenic scores explain a small fraction of the variation and are not diagnosed. Percentages against 633 people from the reference population.
- The loci scores of GWAS Catalog (sRAGE, arterial stiffness, disc hernia) have few loci and low reliability.
- The genes of rare connective tissue diseases (FBN1, COL3A1, COL5A1 in forms of Marfan and Ehlers-Danlos) cannot be evaluated with a chip: the absence of signal here does not exclude these diseases.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001854 (PGS Catalog; Penalized regression; Inguinal hernia; PGP000263, PMID 34995502 · PGS001845 (PGS Catalog; Penalized regression; Varicose vein; PGP000263, PMID 34995502 · PGS001880 (PGS Catalog; Penalized regression; Contracture of palmar fascia [Dupuytren's disease]; PGP000263, PMID 34995502 · PGS001953 (PGS Catalog; Penalized regression; Glycated haemoglobin (HbA1c); PGP000263, PMID 34995502 · PGS000753 (PGS Catalog; Pruning and Thresholding (P+T); Abdominal aortic aneurysm; PGP000159, PMID 32981348).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

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Generated in 02-10-2026 (local time) by the scripts of this report.